

Abhilash Reddy

Senior Software Engineer | Applied AI | AI Platforms | Full Stack

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SUMMARY

Senior Software Engineer with 5+ years of experience designing, building, and deploying enterprise software, AI-powered applications, workflow automation platforms, and production backend systems. Experienced in delivering end-to-end solutions spanning requirements analysis, system design, API development, frontend engineering, backend services, cloud integrations, and production support.

Hands-on experience building Generative AI applications, Retrieval-Augmented Generation (RAG) systems, agentic workflows, LLM-powered platforms, AI orchestration services, and production-grade automation solutions. Strong foundation in combining deterministic software engineering with probabilistic AI through structured outputs, tool calling, evaluation, observability, caching, and reliability engineering.

Proficient in Python, FastAPI, Node.js, Angular, TypeScript, Azure AI, LangChain, LangGraph, REST APIs, and modern software engineering practices, with experience translating complex business requirements into scalable, reliable, production-ready AI platforms.

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102) [Credential]

Microsoft Foundry, Azure OpenAI, Azure AI Search, Document Intelligence, Content Safety, Agentic AI Solutions 2026

EXPERIENCE

Bajaj Finserv (Payroll: Snapwork Technologies)

Senior Software Engineer

Pune, India

Jul 2025 – Present

- Design and develop enterprise backend services, REST APIs, workflow orchestration platforms, and AI-powered automation solutions supporting large-scale financial and loan-processing systems using Python, FastAPI, Azure AI Services, and event-driven architecture.
- Build Applied AI capabilities integrating Azure OpenAI, LLMs, prompt engineering, structured outputs, tool calling, and business-rule engines to deliver reliable decision-support workflows while maintaining deterministic validation and auditability.
- Develop scalable orchestration services integrating internal platforms, business workflows, external APIs, asynchronous processing, and AI services with a strong focus on modular architecture, reusability, and production reliability.
- Implement production-grade engineering practices including observability, distributed tracing, structured logging, retries, caching, schema validation, exception handling, monitoring, and performance optimization across mission-critical workflows.
- Collaborate with business stakeholders, product owners, architects, QA teams, and engineering teams to translate complex financial requirements into scalable software solutions and AI-enabled business workflows deployed in production.

CloudBolt Software

Software Engineer

Pune, India

Mar 2023 – Jun 2025

- Designed and developed enterprise cloud management capabilities spanning frontend applications, backend services, REST APIs, workflow automation, RBAC, distributed systems, and cloud platform integrations.
- Built AI-assisted knowledge retrieval solutions leveraging Retrieval-Augmented Generation (RAG), semantic search, vector search, metadata-aware retrieval, embeddings, and LLM-powered contextual response generation for internal developer platforms.
- Engineered scalable document ingestion, chunking, indexing, hybrid retrieval, relevance optimization, and contextual information pipelines to improve grounded AI responses and developer productivity.
- Integrated enterprise authentication, authorization, cloud infrastructure, and internal platform APIs while improving scalability, application performance, usability, and operational visibility.
- Collaborated with product managers, architects, QA engineers, and cross-functional teams to design, implement, test, and deploy production-ready enterprise software enhancements.

Oerlikon Balzers

Software Engineer

Pune, India

May 2021 – Mar 2023

- Developed enterprise web applications using Angular, TypeScript, RxJS, REST APIs, reusable component-based architecture, and responsive UI design supporting global manufacturing operations.
- Designed scalable frontend architecture with reactive programming, lazy loading, modular components, state management, backend integrations, and performance optimization.
- Worked closely with business users and technical teams to analyze requirements, design software solutions, implement new features, and deliver production-ready releases.

- Improved software quality through automated testing, structured validation, reusable architecture, code reviews, debugging, and continuous enhancement of enterprise applications.

Oerlikon Balzers

Business Analyst

India & Philippines

Sep 2017 – May 2021

- Partnered with business stakeholders, manufacturing teams, and leadership to analyze business processes, identify digital transformation opportunities, and drive workflow optimization initiatives.
- Gathered and documented business requirements, prepared functional specifications, coordinated cross-functional stakeholders, and supported successful delivery of enterprise software projects.
- Performed KPI analysis, CRM reporting, process mapping, business process improvement, operational analysis, and data-driven decision support across multiple international business units.
- Acted as the bridge between business and engineering teams, translating customer requirements into scalable software solutions while ensuring alignment with operational objectives.

APPLIED AI SYSTEMS

Career Workflow — Python, FastAPI, React 19, Playwright, SQLite, LLMs [GitHub]

- * Built a closed-loop AI job operations platform that intelligently discovers, evaluates, ranks, and executes job applications using candidate-aware reasoning, policy controls, and outcome-driven workflow orchestration.
- * Designed an intelligent inference and decision layer combining deterministic rules, LLM-based evaluation, multi-stage filtering, caching, and adaptive strategy to reduce unnecessary model calls while improving application quality.
- * Engineered a production-oriented FastAPI backend with multi-provider acquisition, persistent decision state, runtime recovery, lifecycle tracking, and policy-controlled application execution across complex workflows.
- * Built a React 19 Operations Console for pipeline control, analytics, decision inspection, system diagnostics, and end-to-end operational visibility across the autonomous job workflow.

Pramya — Python, FastAPI, React 19, LangGraph, PostgreSQL, pgvector, DeepSeek [GitHub]

- * Built an evidence-driven AI career workspace that transforms resumes, job descriptions, and interview performance into persistent career intelligence, deterministic readiness assessment, and adaptive interview preparation.
- * Engineered grounded adaptive interviews across text and live voice using LangGraph, RAG, DeepSeek, local speech models, WebSockets, and persistent preparation memory that continuously informs subsequent sessions.

AI Customer Support Copilot — Python, FastAPI, LangGraph, Azure OpenAI, RAG [GitHub]

- * Developed an enterprise AI customer support platform integrating hybrid RAG, LangGraph agent workflows, Azure OpenAI, structured outputs, tool calling, and interchangeable local/cloud LLM providers.
- * Implemented advanced retrieval pipelines with hybrid search, parent-child chunking, cross-encoder reranking, semantic caching, streaming responses, evaluation workflows, and production observability.

RAG API Assistant — Python, FastAPI, FAISS, BM25, LangChain [GitHub]

- * Engineered a modular Retrieval-Augmented Generation platform supporting document ingestion, hybrid retrieval, vector search, reranking, contextual compression, and grounded response generation for developer documentation.
- * Built reusable ingestion, embedding, evaluation, and REST API layers enabling rapid experimentation with retrieval strategies, embedding models, and LLM providers.

TECHNICAL SKILLS

Generative AI: LLMs, RAG, Advanced RAG, Agentic AI, AI Agents, Prompt Engineering, Context Engineering, Tool Calling, Structured Outputs, Function Calling, AI Evaluation

AI Frameworks: LangChain, LangGraph, AutoGen, Semantic Kernel, Hugging Face Transformers, Sentence Transformers

Retrieval: Hybrid Search, Vector Search, BM25, FAISS, Embeddings, Semantic Search, Semantic Ranking, Reranking, Parent-Child Retrieval, Multi-Query Retrieval

Azure AI: Microsoft Foundry, Azure OpenAI, Azure AI Search, Foundry Agent Service, Azure AI Evaluation, Azure AI Services, Application Insights, Responsible AI

Backend: Python, FastAPI, Node.js, Express.js, REST APIs, MongoDB, Redis, Celery, Async Processing, SSE

Frontend: Angular (v8–19), RxJS, TypeScript, JavaScript, HTML5, CSS3

Production Engineering: Docker, CI/CD, Git, Caching, Observability, OpenTelemetry, Distributed Tracing, Retries, Rate Limiting, Circuit Breakers, Pytest, Jest

EDUCATION

M.E. in Mechanical Design Engineering

Savitribai Phule Pune University

Pune, Maharashtra

B.E. in Mechanical Engineering

Savitribai Phule Pune University

Pune, Maharashtra